Design Proposal: Comparative Tolerance and Group Index Analysis of Quasi-TE and Quasi-TM Silicon Strip Waveguides Using Mach-Zehnder Interferometers

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I. Introduction

The primary objective of this design proposal is to create an integrated photonic circuit featuring asymmetric Mach-Zehnder Interferometers (MZIs) to experimentally extract the waveguide group index (n_g) for both quasi-TE and quasi-TM polarizations.

Furthermore, this project investigates fabrication tolerances by implementing controlled variations in waveguide width (480 nm, 500 nm, and 520 nm). This approach allows us to compare how geometric non-idealities during manufacturing affect the spectral response and optical performance of each polarization.

II. Theory and Derivations

The transmission spectrum (transfer function) of an asymmetric Mach-Zehnder Interferometer is governed by the phase difference ($\Delta\phi$) accumulated between its two arms due to the path length imbalance ($\Delta L = L_2 - L_1$). Assuming an ideal 50:50 splitter and combiner, the normalized power transmission $T(\lambda)$ is expressed mathematically as:

$$T(\lambda) = \cos^2\left(\frac{\pi \cdot n_{\text{eff}}(\lambda) \cdot \Delta L}{\lambda}\right) \tag{1}$$

Where $n_{\text{eff}}(\lambda)$ is the wavelength-dependent effective refractive index of the waveguide mode, and λ is the operating wavelength.

The Free Spectral Range (FSR) is defined as the distance between adjacent interference maxima in the transmission spectrum. To derive the relationship for the group index (n_g), we consider the phase condition for two consecutive resonant peaks:

$$\Delta\phi(\lambda_1) - \Delta\phi(\lambda_2) = 2\pi \tag{2}$$

Applying a Taylor series expansion for $n_{\text{eff}}(\lambda)$ around the central wavelength λ_0 and utilizing the definition of the group index $n_g = n_{\text{eff}} - \lambda_0 \frac{dn_{\text{eff}}}{d\lambda}$, we obtain the final expression used to experimentally extract the waveguide group index from the measured FSR data:

$$n_g = \frac{\lambda_0^2}{\text{FSR} \cdot \Delta L} \tag{3}$$

III. Parameter Variations

The proposed variations for the integrated Mach-Zehnder Interferometers are summarized in Table I.

TABLE I: MZI Design Parameter Variations

Design ID	Width	Radius	ΔL	Target FSR
MZI_TE_500	500 nm	10 μm	40 μm	$\sim$14.3 nm
MZI_TE_480	480 nm	10 μm	40 μm	$\sim$14.1 nm
MZI_TE_520	520 nm	10 μm	40 μm	$\sim$14.5 nm
MZI_TM_500	500 nm	30 μm	45 μm	$\sim$14.8 nm
MZI_TM_480	480 nm	30 μm	45 μm	$\sim$14.4 nm
MZI_TM_520	520 nm	30 μm	45 μm	$\sim$15.2 nm

IV. Design Justification

The choices of ΔL (40 μm and 45 μm) ensure that the resulting FSR is around 14–15 nm, which is safely below the maximum measurement bandwidth restriction of 50 nm, preventing aliasing during testing.

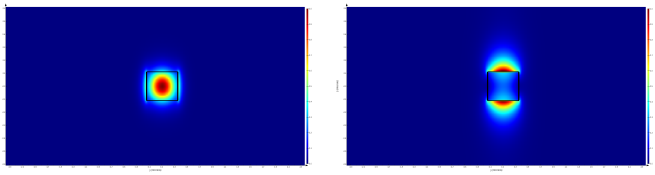
Crucially, the bend radius for the quasi-TM designs is fixed at 30 μm to restrict total bending losses below the tight 0.1% threshold, as verified by numerical eigensolver simulations. For quasi-TE designs, a standard 10 μm radius is sufficient.

V. Waveguide Geometry and Mode Profiles

The integrated circuit utilizes silicon-on-insulator (SOI) strip waveguides with a fixed fabrication height of 220 nm. The core geometry features an initial target width of 500 nm, surrounded by a silicon dioxide (SiO_2) cladding layer.

To satisfy the stringent design objective of maintaining total bending losses below 0.1%, numerical mode analysis dictates distinct bend radii for the two supported polarizations. The quasi-TE mode operates efficiently with a standard bend radius of 10 μm, while the quasi-TM mode requires an expanded bend radius of 30 μm to suppress radiation leakage into the cladding.

The cross-sectional electric field profiles for both fundamental modes at the central operating wavelength of 1.55 μm were simulated using Lumerical MODE Solutions and are illustrated in Fig. 1.



(a) Quasi-TE Mode (b) Quasi-TM Mode

Fig. 1: Simulated fundamental electric field profiles from Lumerical MODE for: (a) Quasi-TE mode with horizontal confinement, and (b) Quasi-TM mode showcasing vertical field discontinuities at the horizontal boundaries.

VI. Compact Dispersion Models

To establish an accurate circuit simulation framework in Lumerical INTERCONNECT, compact numerical models representing the wavelength-dependent effective refractive index (n_{eff}) were extracted using a 2nd-order polynomial fit focused around the central wavelength of 1.55 μm.

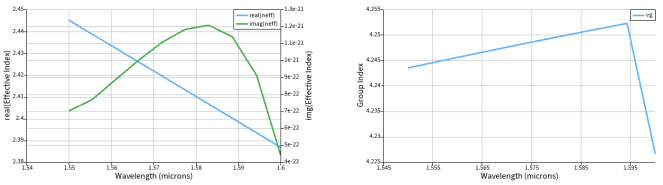
For the fundamental Quasi-TE mode, the mathematical compact expression is given by:

$$n_{\text{eff, TE}}(\lambda) = 2.4455 - 1.16205(\lambda - 1.55) - 0.07572(\lambda - 1.55)^2 \tag{4}$$

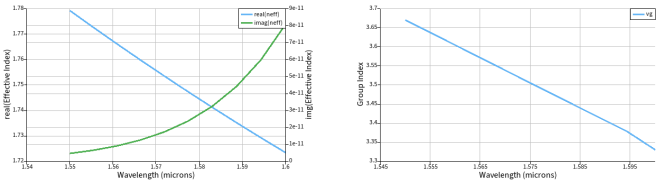
For the fundamental Quasi-TM mode, the corresponding polynomial extraction yields:

$$n_{\text{eff, TM}}(\lambda) = 1.7842 - 1.18610(\lambda - 1.55) - 0.0832(\lambda - 1.55)^2 \tag{5}$$

The continuous spectral dependency of both the effective index (n_{eff}) and the resulting group index (n_g) plotted against the operating wavelength band is presented in Fig. 2.



(a) TE n_{eff} (b) TE n_g



(c) TM n_{eff} (d) TM n_g

Fig. 2: Waveguide dispersion curves simulated across the 1.50 μm to 1.60 μm sweep bandwidth: (a) Quasi-TE effective index, (b) Quasi-TE group index (∼4.2), (c) Quasi-TM effective index, and (d) Quasi-TM group index (∼3.6).

VII. Circuit Simulation and Transmission Spectra

The complete MZI architectures with asymmetric arm imbalances ($\Delta L_{\text{TE}} = 40$ μm and $\Delta L_{\text{TM}} = 45$ μm) were implemented and analyzed within Lumerical INTERCONNECT. An Optical Network Analyzer (ONA) block was utilized to record the normalized transmission spectral signatures across a 50 nm measurement window.

The resulting high-fidelity interference fringes, showcasing stable sinusoidal oscillations, are captured in Fig. 3. The variations in waveguide width alter the effective index, causing a controlled phase shift and spectral translation of the resonant peaks, which directly quantifies the geometric fabrication tolerances.

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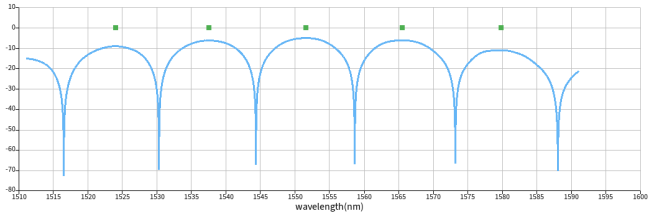
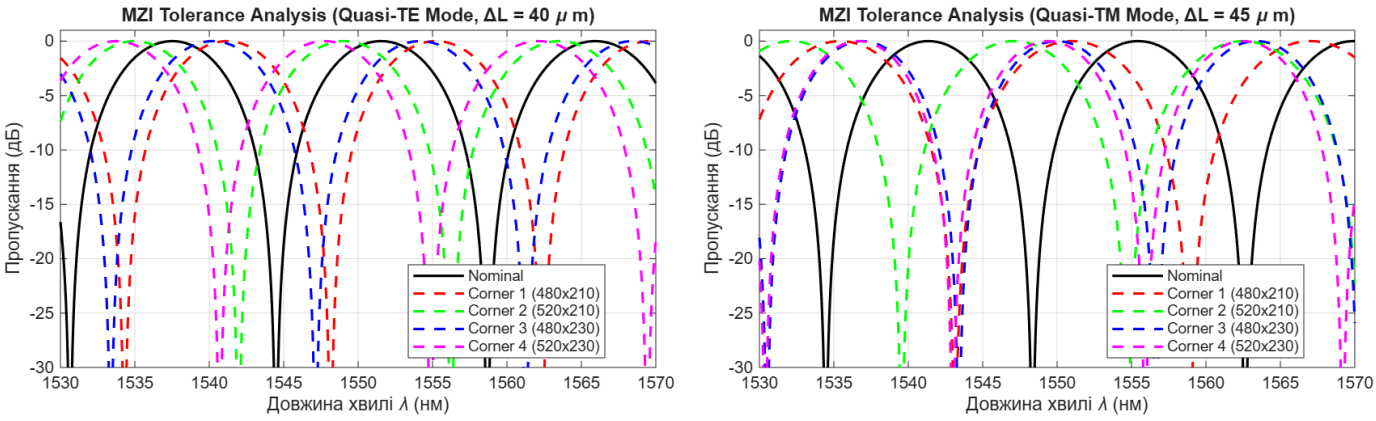


Fig. 3: Normalized transmission spectrum of the asymmetric Mach-Zehnder Interferometer (MZI) compiled in Lumerical INTERCONNECT for the Quasi-TE polarization, showcasing characteristic sinusoidal interference fringes and extracted Free Spectral Range (FSR) resonant peaks.

IX. Fabrication Tolerance and Corner Analysis

To evaluate the robustness of the designed asymmetric Mach-Zehnder Interferometers against manufacturing non-uniformities, a 4-corners and 1-nominal design tolerance analysis was performed. Geometric variations in the waveguide cross-section were introduced, considering an industrial lithographic tolerance of $\Delta w = \pm 20$ nm for the waveguide width and an etching tolerance of $\Delta h = \pm 10$ nm for the silicon core height.

The transverse cross-sections for all five structural combinations were simulated using the 2D Finite Difference Eigenmode (FDE) solver in Lumerical MODE, assuming a fully symmetric SiO_2 top and bottom cladding. For each corner case, the exact effective refractive index (n_{eff}) and group index (n_{g}) were extracted at the central operating wavelength of $\lambda_0 = 1.55\,\mu$m. The extracted parameters for both the fundamental Quasi-TE and Quasi-TM modes are summarized in Table **??**.



(a) Quasi-TE mode transmission ($\Delta L = 40\,\mu$m)

(b) Quasi-TM mode transmission ($\Delta L = 45\,\mu$m)

Fig. 4: MZI fabrication tolerance analysis (4 corners and 1 nominal design) for both fundamental modes under variations of ± 20 nm in width and ± 10 nm in height.

A. Extracted Tolerance Parameters and FSR Analysis

The numerical results extracted from the 2D FDE simulations for both polarizations are organized into individual compact models. Table II summarizes the parameters for the fundamental Quasi-TE mode, while Table III presents the corresponding analysis for the Quasi-TM mode.

TABLE II: Quasi-TE Mode Corner Analysis Parameters ($\Delta L = 40\,\mu$m)

Design Case	n_{eff}	n_g	FSR (nm)
Nominal (500/220 nm)	2.445454	4.241060	14.16
Corner 1 (480/210 nm)	2.378048	4.265354	14.08
Corner 2 (520/210 nm)	2.438712	4.200967	14.30
Corner 3 (480/230 nm)	2.452717	4.286174	14.01
Corner 4 (520/230 nm)	2.512588	4.218595	14.24

TABLE III: Quasi-TM Mode Corner Analysis Parameters ($\Delta L = 45\,\mu$m)

Design Case	n_{eff}	n_g	FSR (nm)
Nominal (500/220 nm)	1.769860	3.778210	14.13
Corner 1 (480/210 nm)	1.689925	3.387083	15.76
Corner 2 (520/210 nm)	1.715648	3.496593	15.27
Corner 3 (480/230 nm)	1.825480	4.034492	13.23
Corner 4 (520/230 nm)	1.858436	4.124905	12.94

Using the extracted modal data, the continuous transmission spectra across the C-band (1530–1570 nm) were analytically modeled in MATLAB, incorporating full waveguide dispersion. As illustrated in Fig. 4, the fabrication tolerances induce a deterministic spectral translation of the resonant peaks (fringe shift) relative to the nominal response. The Quasi-TM mode exhibits higher sensitivity to vertical scale drift (Δh), while the Quasi-TE mode tracks lateral variations (Δw), validating the necessity of a well-optimized top oxide cladding for phase-stable photonic circuit operation.

X. Compact Dispersion Models

To establish an accurate photonic circuit simulation framework, compact numerical models representing the wavelength-dependent effective refractive index $n_{\text{eff}}(\lambda)$ were extracted using a 2nd-order Taylor polynomial expansion. The analytical representation is formulated around the central operating wavelength of $\lambda_0 = 1.55\,\mu$m as follows:

$$n_{\text{eff}}(\lambda) \approx n_{\text{eff0}} - a_1(\lambda - 1.55) - a_2(\lambda - 1.55)^2 \tag{6}$$

where the linear and quadratic dispersion coefficients are calculated from the simulated modal indices as $a_1 = (n_g - n_{\text{eff0}})/\lambda_0$ and $a_2 = (n_g - n_{\text{eff0}})/\lambda_0^2$.

A. Quasi-TE Fundamental Mode Models

Applying the curve fitting methodology to the extracted data for the Quasi-TE polarization (Mode 1), the independent compact models for all five simulation cases are defined below:

- **Nominal Design (500 / 220 nm):**

$$n_{\text{eff, TE_Nom}}(\lambda) \approx 2.4455 - 1.1585(\lambda - 1.55) - 0.7474(\lambda - 1.55)^2 \tag{7}$$

- **Corner 1 (480 / 210 nm, Narrow-Thin):**

$$n_{\text{eff, TE_C1}}(\lambda) \approx 2.3780 - 1.2176(\lambda - 1.55) - 0.7856(\lambda - 1.55)^2 \tag{8}$$

- **Corner 2 (520 / 210 nm, Wide-Thin):**

$$n_{\text{eff, TE_C2}}(\lambda) \approx 2.4387 - 1.1370(\lambda - 1.55) - 0.7335(\lambda - 1.55)^2 \tag{9}$$

- **Corner 3 (480 / 230 nm, Narrow-Thick):**

$$n_{\text{eff, TE_C3}}(\lambda) \approx 2.4527 - 1.1829(\lambda - 1.55) - 0.7631(\lambda - 1.55)^2 \tag{10}$$

- **Corner 4 (520 / 230 nm, Wide-Thick):**

$$n_{\text{eff, TE_C4}}(\lambda) \approx 2.5126 - 1.1006(\lambda - 1.55) - 0.7101(\lambda - 1.55)^2 \tag{11}$$

B. Quasi-TM Fundamental Mode Models

Similarly, the extracted models representing the corresponding fabrication tolerance variations for the Quasi-TM polarization (Mode 2) yield:

- **Nominal Design (500 / 220 nm):**

$$n_{\text{eff, TM_Nom}}(\lambda) \approx 1.7699 - 1.2957(\lambda - 1.55) - 0.8359(\lambda - 1.55)^2 \tag{12}$$

- **Corner 1 (480 / 210 nm, Narrow-Thin):**

$$n_{\text{eff, TM_C1}}(\lambda) \approx 1.6899 - 1.0949(\lambda - 1.55) - 0.7064(\lambda - 1.55)^2 \quad (13)$$

- **Corner 2 (520 / 210 nm, Wide-Thin):**

$$n_{\text{eff, TM_C2}}(\lambda) \approx 1.7156 - 1.1490(\lambda - 1.55) - 0.7413(\lambda - 1.55)^2 \quad (14)$$

- **Corner 3 (480 / 230 nm, Narrow-Thick):**

$$n_{\text{eff, TM_C3}}(\lambda) \approx 1.8255 - 1.4251(\lambda - 1.55) - 0.9195(\lambda - 1.55)^2 \quad (15)$$

- **Corner 4 (520 / 230 nm, Wide-Thick):**

$$n_{\text{eff, TM_C4}}(\lambda) \approx 1.8584 - 1.4620(\lambda - 1.55) - 0.9432(\lambda - 1.55) \quad (16)$$

C. Monte Carlo Tolerance Analysis

To evaluate the impact of fabrication imperfections on the device performance, a Monte Carlo tolerance analysis was performed. This simulation specifically accounts for wafer-to-wafer variations under the Electron Beam Lithography (Applied Nanotools) technology.

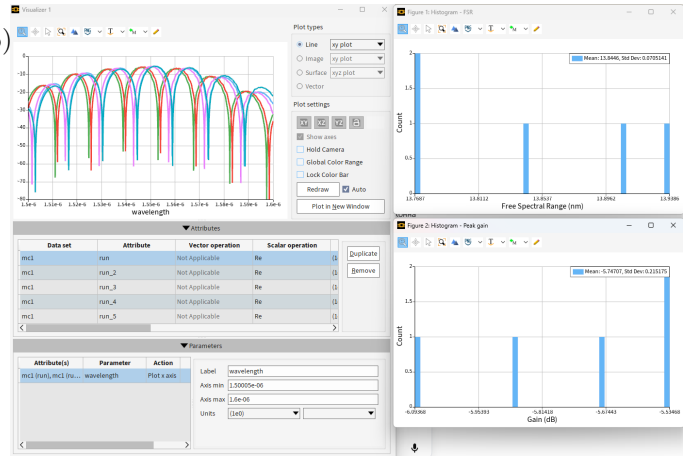
The simulation parameters and the corresponding geometrical variations are summarized in Table IV.

TABLE IV: Monte Carlo Simulation Parameters for Wafer-to-Wafer Variation.

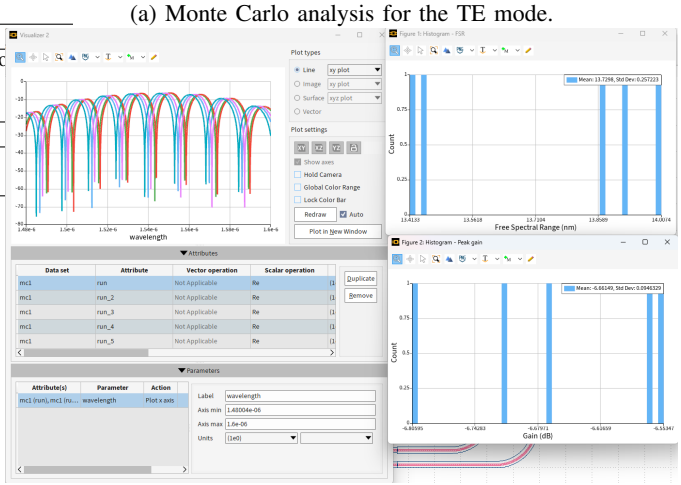
Simulation Parameter	**Value**
Fabrication Technology	Electron Beam Lithography (Applied Nanc
Number of Test Wafers	1
Number of Test Dies Per Wafer	5
Geometrical Variation	**Std. Dev. (σ)**
Width Variation	5.0 nm
Thickness Variation	3.0 nm

The combined statistical results of the Monte Carlo simulations, including the spectral responses and the extracted parameter histograms (FSR and Peak Gain), are presented in Fig. 5 for both TE and TM modes.

The extracted statistical indicators from the generated histograms demonstrate the robustness of the device against fabrication tolerances. For the TE mode, the Free Spectral Range (FSR) exhibits a mean value (μ) of approximately 13.84 nm with a standard deviation (σ) of 0.07 nm, while the Peak Gain shows $\mu \approx -5.75$ dB and $\sigma \approx 0.21$ dB. For the TM mode, the FSR demonstrates $\mu \approx 13.73 filename$ nm ($\sigma \approx 0.26$ nm) and the Peak Gain shows $\mu \approx -6.66$ dB ($\sigma \approx 0.09$ dB).



(a) Monte Carlo analysis for the TE mode.



(b) Monte Carlo analysis for the TM mode.

Fig. 5: Wafer-to-wafer Monte Carlo simulation results showing spectral variations and extracted histograms (FSR and Peak Gain) for: (a) TE mode, and (b) TM mode.